

Array Range Queries

1 Introduction

Array range queries are used to compute values over a subarray efficiently. Instead of re-computing values for every query, preprocessing techniques are applied.

Let the array be:

$$A = [3, 2, 4, 5, 1, 6, 7, 2]$$

2 Range Sum Query (RSQ)

2.1 Definition

Find the sum of elements between indices L and R .

$$\text{Sum}(L, R) = \begin{cases} P[R], & L = 0 \\ P[R] - P[L - 1], & L > 0 \end{cases}$$

2.2 Prefix Sum Array

$$P = [3, 5, 9, 14, 15, 21, 28, 30]$$

2.3 Example

$$\text{Sum}(2, 5) = 21 - 5 = 16$$

Time Complexity: $O(1)$ per query

3 Range Minimum Query (RMQ)

3.1 Definition

Find the minimum element in the range $[L, R]$.

3.2 Naive Approach

$$\min(A[L \dots R])$$

3.3 Example

$$\text{Min}(2, 5) = \min(4, 5, 1, 6) = 1$$

Optimized Methods:

- Sparse Table: $O(1)$ query
- Segment Tree: $O(\log n)$

4 Range Maximum Query

4.1 Definition

Find the maximum element in the range $[L, R]$.

4.2 Example

$$\text{Max}(2, 5) = \max(4, 5, 1, 6) = 6$$

Optimized Methods:

- Segment Tree
- Sparse Table

5 Range Average Query

5.1 Definition

Find the average of elements in range $[L, R]$.

$$\text{Average}(L, R) = \frac{\text{Sum}(L, R)}{R - L + 1}$$

5.2 Example

$$\text{Average}(2, 5) = \frac{16}{4} = 4$$

Time Complexity: $O(1)$ using prefix sums

6 Range XOR Query

6.1 Definition

Find XOR of elements in the range $[L, R]$.

6.2 Prefix XOR Array

$$X[i] = A[0] \oplus A[1] \oplus \cdots \oplus A[i]$$

$$X = [3, 1, 5, 0, 1, 7, 0, 2]$$

6.3 Query Formula

$$\text{XOR}(L, R) = \begin{cases} X[R], & L = 0 \\ X[R] \oplus X[L - 1], & L > 0 \end{cases}$$

6.4 Example

$$\text{XOR}(2, 5) = 7 \oplus 1 = 6$$

7 Range Frequency Query

7.1 Definition

Count occurrences of a value k in range $[L, R]$.

7.2 Example

$$\text{Frequency of } 2 \text{ in } [0, 7] = 2$$

Optimized Approaches:

- Prefix frequency array
- Hash map with index lists

8 Comparison of Range Queries

Query Type	Preprocessing	Query Time
Sum	Prefix Sum	$O(1)$
Min / Max	Sparse Table	$O(1)$
Average	Prefix Sum	$O(1)$
XOR	Prefix XOR	$O(1)$
Frequency	Prefix Count	$O(1)$

9 Applications

- Competitive programming
- Database analytics
- Financial analysis
- Time-series processing

10 Conclusion

Range query techniques significantly improve performance by avoiding repeated computation. Choosing the correct preprocessing method depends on query type and constraints.